PHY-340 Homework Set 2

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Problem 1

We have a 20 keV deuteron at the midplane of a mirror fusion device with $\alpha = 45^\circ$, $B = 0.7\,\mathrm{T}$. Recall the definition of the Larmor radius,

$$r_L = \frac{mv_\perp}{qB} \tag{1.1}$$

where $v_\perp$ is the velocity component in the direction perpendicular to the magnetic field. Since α is the angle between the velocity and the parallel direction, we know that $v_\perp = v \sin\alpha$. In terms of kinetic energy, $v = \sqrt{2K/m}$, so

$$v_\perp = \sin\alpha\sqrt{\frac{2K}{m}}. \tag{1.2}$$

Plugging this relation back into (1.1), we get

$$r_L = \frac{m\sin\alpha\sqrt{2K/m}}{qB} = \frac{\sin\alpha\sqrt{2Km}}{eB} \tag{1.3}$$

$$= \frac{\sqrt{2}}{2}\frac{\sqrt{2(3.2\times10^{-15}\,\mathrm{J})(2\times1.67\times10^{-27}\,\mathrm{kg})}}{(1.602\times10^{-19}\,\mathrm{C})(0.7\,\mathrm{T})} \tag{1.4}$$

$$= \sqrt{2}\,\frac{2.312\times10^{-21}}{1.121\times10^{-19}}\,\mathrm{m} \tag{1.5}$$

$$= 29.2\,\mathrm{mm} \tag{1.6}$$

Problem 2

We have a proton moving in uniform electric and magnetic fields given by $\mathbf{E} = 10\,\mathrm{V\,m^{-1}}\,\hat{x}$ and $\mathbf{B} = 10^{-4}\,\mathrm{T}\,\hat{z}$. The proton has an initial velocity of $\mathbf{v}_0 = 2.414\times10^5\,\mathrm{m\,s^{-1}}\,\hat{x} - 10^5\,\mathrm{m\,s^{-1}}\,\hat{y}$ and an initial position of $\mathbf{x} = 0$.

(a) The cyclotron frequency ω_c is given by the formula

$$\omega_c = \frac{|q|B}{m} = \frac{(1.602\times10^{-19}\,\mathrm{C})(10^{-4}\,\mathrm{T})}{1.672\times10^{-27}\,\mathrm{kg}} = \boxed{9.58\,\mathrm{kHz}} \tag{2.1}$$

(b) To solve for the position $\mathbf{x}(t)$ and the velocity $\mathbf{v}(t)$ as a function of time, we begin with the Lorentz force law,

$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = q\left(\mathbf{E} + \mathbf{v}\times\mathbf{B}\right), \tag{2.2}$$

and simplify using our knowledge that $E_y = E_z = 0$ and $B_x = B_y = 0$,

$$m\dot{v}_x = qE + qv_yB \qquad \text{or} \qquad v_x = \frac{q}{m}E + \omega_c v_y \tag{2.3}$$

$$m\dot{v}_y = -qv_xB \qquad \text{or} \qquad v_y = -\omega_c v_x \tag{2.4}$$

$$m\dot{v}_z = 0. \tag{2.5}$$

In order to solve the differential equations for v_x and v_y, we start by differentiating each equation, to separate the x and y components,

$$\ddot{v}_x = \omega_c \dot{v}_y = -\omega_c^2 v_x \tag{2.6}$$

and

$$\ddot{v}_y = -\omega_c \dot{v}_x = -\omega_c^2 \left(\frac{E}{B} + v_y\right). \tag{2.7}$$

Notice that these equations imply a complex exponential solution. We solve the x-component as

$$v_x(t) = \mathrm{Re}\left(v_\perp e^{i(\omega_c t + \delta)}\right) = v_\perp \cos(\omega_c t + \delta). \tag{2.8}$$

We solve the y-component as

$$v_y(t) = \mathrm{Re}\left(-\frac{E}{B} + v_\perp e^{i(\omega_c t + \delta)})\right) = -\frac{E}{B} + v_\perp \sin(\omega_c t + \delta) \tag{2.9}$$

Finally, we must take into account the initial conditions, $v_{x0} = 2.414 \times 10^5\,\mathrm{m\,s^{-1}}$ and $v_{y0} = -10^5\,\mathrm{m\,s^{-1}}$. We let $\delta = 0$ and notice that $-\frac{E}{B} = -10^5\,\mathrm{m\,s^{-1}}$, precisely the initial y-velocity, so

$$\boxed{v_y(t) = -\frac{E}{B} + v_\perp \sin(\omega_c t)} \tag{2.10}$$

$$\boxed{v_x(t) = v_\perp \cos(\omega_c t)} \tag{2.11}$$

Where $v_\perp = 2.414 \times 10^5\,\mathrm{m\,s^{-1}}$, the initial x-velocity. Integrating these two equations and recalling that $x_0 = y_0 = 0$, we get the positions

$$\boxed{x(t) = r_L \sin(\omega_c t)} \tag{2.12}$$

$$\boxed{y(t) = -\frac{E}{B}t - r_L \cos(\omega_c t) + C} \tag{2.13}$$

where $C = r_L$ is the constant of integration necessary to have $y_0 = 0$, or $r_L = \frac{v_\perp}{\omega_c} = 25.2\,\mathrm{m}$.

(c) The $\mathbf{E} \times \mathbf{B}$ drift velocity is given by

$$\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2} = \frac{E}{B}(\hat{x} \times \hat{z}) = \boxed{-10^5\,\mathrm{m\,s^{-1}}\ \hat{y},} \tag{2.14}$$

precisely the initial velocity in the y-direction.

Problem 3

We have a magnetic mirror with the magnetic field strength along the axis defined by $B_z = B_0(1 + \alpha^2 z^2)$.

(a) Suppose an electron has a velocity at $z = 0$ given by $v^2 = 3v_{\parallel 0}^2 = 1.5v_{\perp 0}^2$. We know that the total velocity $v^2 = v_{\perp 0}^2 + v_{\parallel 0}^2$ must be constant (by conservation of energy), and that the electron will be reflected when $v_\parallel = 0$. Thus, reflection will occur where $v_\perp = v$, or

$$v^2 = \left(\frac{B_z}{B_0}\right) v_{\perp 0}^2. \tag{3.1}$$

Substituting our definition of B_z, we can determine the value of z at which the electron is reflected,

$$v^2 = \left(1 + \alpha^2 z^2\right) v_{\perp 0}^2 \quad \Longrightarrow \quad 1 + \alpha^2 z^2 = \frac{v^2}{v_{\perp 0}^2} = \frac{3}{2} \tag{3.2}$$

$$\alpha^2 z^2 = \frac{1}{2} \tag{3.3}$$

$$\boxed{z = \pm \frac{1}{a}\frac{\sqrt{2}}{2}} \tag{3.4}$$

(b) We start with the Lorentz force law in the direction parallel to the magnetic field,

$$\mathbf{F}_\parallel = m\ddot{z} = q(\mathbf{v} \times \mathbf{B})_z = qv_\perp B_r. \tag{3.5}$$

Cravens (p. 57) provide an expression for B_r (provided $B_r \ll B_z$),

$$B_r \approx -\frac{1}{2} r_L \frac{\mathrm{d}B_z}{\mathrm{d}z}. \tag{3.6}$$

Adopting this relationship, we get the equation of motion

$$m\ddot{z} = -\frac{1}{2} q v_\perp r_L \frac{\mathrm{d}B_z}{\mathrm{d}z}. \tag{3.7}$$

In order to simplify the equation of motion, we must use some key relations. First, recall the definition of the Larmor radius $r_L = \frac{mv_\perp}{qB_z}$. Additionally, recall the relationship $v_\perp^2 = \left(\frac{B_z}{B_0}\right) v_{\perp 0}^2$. Finally, we must differentiate B_z,

$$\frac{\mathrm{d}B_z}{\mathrm{d}z} = B_0 \frac{\mathrm{d}}{\mathrm{d}z}\left(1 + \alpha^2 z^2\right) = 2B_0 \alpha^2 z. \tag{3.8}$$

First, plugging in r_L, our equation of motion becomes

$$m\ddot{z} = -\frac{1}{2}\frac{mv_\perp^2}{B_z}\frac{\mathrm{d}B_z}{\mathrm{d}z}. \tag{3.9}$$

Plugging in $v_\perp^2$, we get

$$\ddot{z} = -\frac{1}{2}\frac{v_{\perp 0}^2}{B_z}\left(\frac{B_z}{B_0}\right)\frac{\mathrm{d}B_z}{\mathrm{d}z} = -\frac{1}{2}\frac{v_{\perp 0}^2}{B_0}\frac{\mathrm{d}B_z}{\mathrm{d}z}. \tag{3.10}$$

Finally, plugging in $\frac{\mathrm{d}B_z}{\mathrm{d}z}$ we get

$$\ddot{z} = -\frac{1}{2}\frac{v_{\perp 0}^2}{B_0}(2B_0\alpha^2 z) \tag{3.11}$$

$$\boxed{\ddot{z} = -\alpha^2 v_{\perp 0}^2 z} \tag{3.12}$$

(c) The equation of motion (3.12) represents simple harmonic motion, as the second derivative $\ddot{z}$ is negatively proportional to the value of z. Thus, the motion is sinusoidal with the frequency $\Omega = \alpha v_{\perp 0}$.

Problem 4 RLS EOC 3.2

(a) Figure 1 shows the guiding center velocity versus the starting perpendicular velocity for a proton in an $\mathbf{E} \times \mathbf{B}$ field. In order to compute the guiding center velocity, I let the simulation run for 1 gyration and noted the time at which it completed its gyration and how far it had moved in the y-direction from its starting point. It is clear from this figure that the guiding center velocity is independent of the starting velocity. This is consistent with our formula for the $\mathbf{E} \times \mathbf{B}$ drift velocity, $\mathbf{v}_E = \frac{\mathbf{E}\times\mathbf{B}}{B^2}$, which depends only on the field strengths and relative directions.

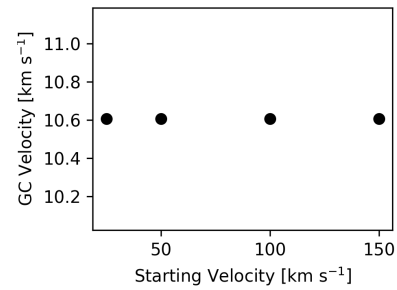

Figure 1: Guiding center velocity vs the starting perpendicular velocity for a proton in cross electric and magnetic fields.

(d) Figure 2 shows how particle mass effects the guiding center velocity. Higher-mass particles appear to have lower guiding center velocities, which does not agree well with theory.

Figure 3 shows the trajectories of both a proton (red) and a negatively charged H- ion (blue, in place of an electron, which are too small in mass to generate a proper trajectory). We see that charge does not affect the guiding center motion, but does cause the two particles to gyrate in opposite directions.

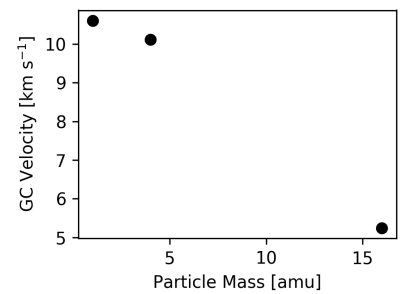

Figure 2: Guiding center velocity vs particle mass.

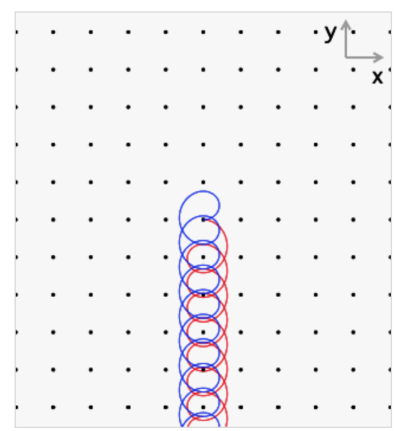

Figure 3: Trajectories of a proton (red) and a H- ion (blue) in crossed electric and magnetic fields.

Problem 5 RLS EOC 3.3

(a) Figure 4 shows the guiding center velocity versus the starting perpendicular velocity for a proton in a $\nabla\mathbf{B}$ field. A least-squares linear fit is applied to the log-log plot, showing a strong relationship between the starting velocity and the drift velocity, unlike in the $\mathbf{E} \times \mathbf{B}$ case.

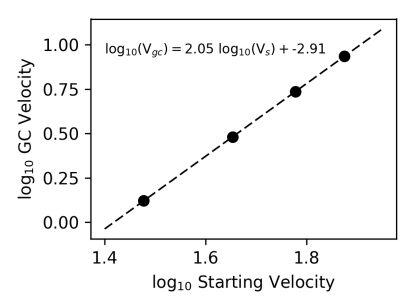

Figure 4: Guiding center velocity vs starting velocity for a proton in a magnetic field gradient.

(d) In a $\nabla \mathbf{B}$ field, electrons and protons drift in opposite directions. Figure 5 shows the trajectories of a proton and an H- ion (in place of an electron).

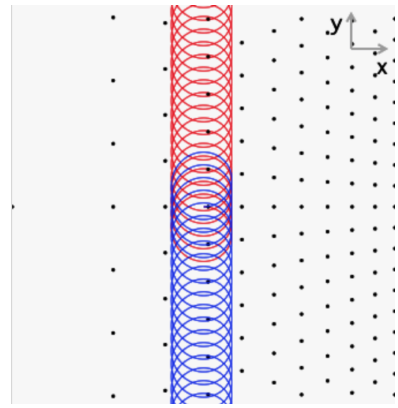

Figure 5: Trajectories of a proton (red) and a H- ion (blue) in a gradient magnetic field.

Problem 6 RLS EOC 3.4

I used the curvature drift component of the particle motion module to first determine that there is no drift motion if the particle is initially given no parallel velocity. I did this by giving the particle only x-velocity, i.e. perpendicular velocity, and noting that it simply gyrated in place and experienced no drift. However, figure 6 shows the trajectories of a proton in a curved magnetic field at three different initial *parallel* velocities. Note that these protons were given *only* parallel velocity. We can see from the figure that a factor of two increase in initial parallel velocity leads to a factor of two increase in the guiding center velocity, i.e. there is a linear relationship between the two speeds.

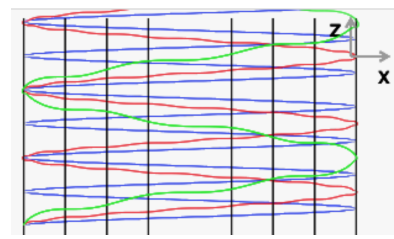

Figure 6: Trajectories of a proton in a curved magnetic field with magnetic field strength $B = 200\,\mathrm{T}$ initial $v_\perp = 0$ and initial $v_\parallel$: $25\,\mathrm{km\,s^{-1}}$ (blue), $50\,\mathrm{km\,s^{-1}}$ (red), and $100\,\mathrm{km\,s^{-1}}$ (green).

Figure 7 shows the trajectories of a proton in a curved magnetic field at fixed initial velocity but three different magnetic field strengths. We see a similar pattern: that a factor of two increase in magnetic field strength leads to a factor of two decrease in gyrovelocity, indicating a negative linear relationship.

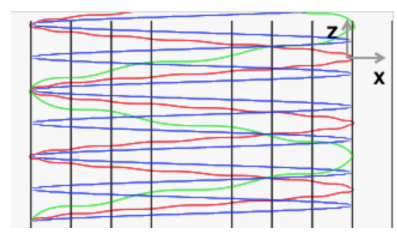

Figure 7: Trajectories of a proton in a curved magnetic field with initial $v_\perp = 0$, initial $v_\parallel = 50\,\mathrm{km\,s^{-1}}$ and magnetic field strength: 400 T (blue), 200 T (red), and 100 T (green).

Figure 8 shows the trajectories of a proton and a H- ion in a curved magnetic field. Note that the two oppositely charged particles drift in opposite directions, so curvature drift does lead to an electric current.

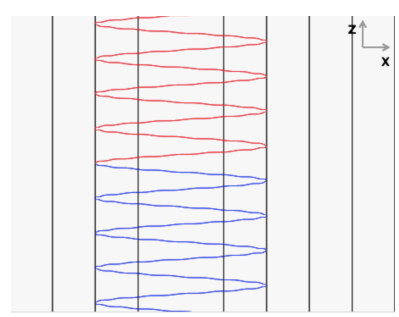

Figure 8: Trajectories of a proton (red) and a H- ion (blue) in a curved magnetic field with nonzero initial parallel velocity.

Note: color figures are available at `hollisakins.github.io/files/HW2.pdf`